



LEARN • BUILD • SHARE • GROW

GitHub

NOTES FOR DATA ANALYSTS

BUILD YOUR PORTFOLIO.
SHOW YOUR SKILLS. **GET HIRED.**

WHAT YOU WILL LEARN



What is GitHub?
Why it is important?



Create Account &
Set up Your Profile



Create Repositories
Like a Professional



Write Effective README
that stands out



Organize Projects
Like a Portfolio



Upload, Commit &
Share Your Work



Best Practices &
Pro Tips



Excel
Projects



SQL
Queries



Power BI
Dashboards



Python
Notebooks

“ Your GitHub is more
than a profile,
it's your **professional story.** ”

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PRACTICAL
LEARNING



REAL WORLD
PROJECTS



CAREER
GROWTH



Bonus Lecture – GitHub for Data Analysts

From Learning Excel to Building Your Professional Portfolio

Module 1 – What is GitHub?

Definition

GitHub is a cloud-based platform that allows developers, data analysts, and professionals to store, manage, share, and showcase their projects.

Simply put,

GitHub is your online portfolio where recruiters can see your work instead of only reading your resume.

Why Should a Data Analyst Learn GitHub?

Many freshers think GitHub is only for programmers.

That is incorrect.

Today companies also expect Data Analysts to maintain their work professionally.

You can upload

- Excel Projects
- Power BI Dashboards
- SQL Scripts
- Python Projects
- Datasets
- Documentation
- Resume
- Certificates

Everything can be organized in one place.

Module 2 – Benefits of GitHub

Professional Portfolio

Recruiters can directly see your work.

Version Control

Never lose previous project versions.

Free Cloud Storage

Projects remain online.

Resume Enhancement

Instead of writing

"I know Excel"

Show

An Excel Dashboard

Practice Workbook

Business Project

Easy Sharing



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One link can showcase your complete portfolio.

Builds Credibility

Projects speak louder than certificates.

Module 3 – Git vs GitHub

Many students confuse these.

Git

Software used to track changes.

Installed on computer.

GitHub

Website

Stores Git repositories online.

Simple analogy

Git

↓

Notebook

GitHub

↓

Cloud Storage

Module 4 – Creating GitHub Account

Steps

1. Visit github.com
2. Sign Up
3. Verify Email
4. Choose Username
5. Complete Profile

Module 5 – Profile Optimization

Professional Profile Photo

Professional Username

Bio

Location

LinkedIn

Portfolio

Email

Example Bio

Data Analyst

Excel | SQL | Power BI | Python

Learning Analytics Everyday



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Building Real Business Projects

Module 6 – What is Repository?

Definition

A Repository (Repo)
is a folder containing one complete project.

Example

Excel Dashboard

↓

One Repository

Power BI Sales Dashboard

↓

Another Repository

SQL Project

↓

Another Repository

Each project

=

One Repository

Module 7 – Creating Repository

Click

New Repository

↓

Repository Name

↓

Description

↓

Public

↓

Initialize README

↓

Create

Module 8 – Repository Naming

Good

Excel-Sales-Dashboard

HR-Analytics-Dashboard

SQL-Employee-Analysis



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PowerBI-Sales-Insights

Excel-Data-Cleaning

Avoid
Project1

New Folder

Practice

Final Final

abc123

Module 9 – What is README?

README

=

Project Documentation

It tells

What project is

Why it was built

Dataset

Skills Used

Business Problem

Screenshots

Results

Recruiters first read README.

Module 10 – README Structure

Project Name

About Project

Business Problem

Dataset

Tools Used

- Excel
- SQL
- Power BI

Features

Dashboard

Business Insights

Files Included

Learning Outcomes



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Module 11 – Uploading Files

Upload

Excel Workbook

Dataset

Images

PDF

Presentation

Documentation

Module 12 – Adding Screenshots

Always upload

Dashboard Screenshot

Before/After Cleaning

Pivot Table

Charts

KPIs

Recruiters like visuals.

Module 13 – Organizing GitHub

Example

GitHub Portfolio

|

|— Excel Dashboard

|— Excel Data Cleaning

|— SQL Projects

|— Power BI Dashboard

|— Python Projects

|— Resume

|— Certificates

|— Practice Workbooks

Module 14 – GitHub Profile README

GitHub allows creating a special profile README.

It becomes your homepage.

Include

Introduction



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Skills
Projects
Contact
Social Links
Badges
Statistics

Module 15 – Making GitHub Attractive

Include
Professional Banner
Project Images
Icons
Badges
Pinned Repositories
Consistent Naming
Professional Descriptions

Module 16 – Pinned Repositories

Pin your best work.
Recommended
Excel Dashboard
Power BI Dashboard
SQL Project
Python Project
Portfolio Project



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Module 17 – GitHub for Interviews

Recruiters may ask
Show me your projects.
Instead of explaining,
Open GitHub.
Show
Repository
README
Dashboard
Excel File
Business Insights

Module 18 – Best Practices

- ✓ One repository per project
- ✓ Good README
- ✓ Proper screenshots
- ✓ Clear project names
- ✓ Business-focused descriptions
- ✓ Keep updating regularly

Module 19 – Common Mistakes

- ✗ Uploading random files
 - ✗ No README
 - ✗ No screenshots
 - ✗ Poor naming
 - ✗ Empty repositories
 - ✗ No descriptions
-

Module 20 – Career Advice

Treat GitHub like your **digital resume**.

Every Excel dashboard, SQL query, Power BI report, Python notebook, or analytics project you build should be uploaded.

When someone visits your GitHub profile, they should immediately understand:

- What skills you have
- What projects you've built
- How you solve business problems
- How consistent you are as a learner

Your GitHub should tell your professional story without you saying a word.

Module – README.md for Data Analysts

What is README?

Definition

A **README.md** file is the first document people see when they open your GitHub repository.

It explains your project in a structured way so that anyone—including recruiters, hiring managers, teammates, or clients—can quickly understand:

- What the project is
- Why it was built
- Which tools were used
- How to use it
- What business problem it solves
- What insights were generated

Think of it as the **landing page** of your project.

Why is README Important?

A good README can be the difference between someone closing your repository in 5 seconds or spending several minutes exploring your work.

Benefits

- Makes your project professional
 - Helps recruiters understand your work
 - Improves GitHub portfolio quality
 - Makes projects easy to navigate
 - Documents your learning journey
 - Shows communication skills
-



Why Every Data Analyst Needs README

Suppose you upload an Excel Dashboard.

Without README

Dashboard.xlsx

Sales.csv

Nobody knows

- What dashboard does
- Dataset source
- KPIs
- Business problem
- Insights

With README

Business Problem

Dataset

Dashboard Preview

KPIs

Business Insights

Files Included

Learning Outcomes

Anyone can understand your project.

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What Does ".md" Mean?

README is written in

Markdown Language

Markdown is a lightweight formatting language.

It allows you to create

- Headings
- Lists
- Tables
- Images
- Links
- Code Blocks
- Quotes
- Checklists

without HTML.

Creating README

Method 1

While creating Repository

✓ Add README

GitHub creates it automatically.

Method 2

Inside Repository

Add File



Create New File



Name

README.md

Basic README Structure

Project Title

About Project

Business Problem

Dataset

Tools Used

Project Workflow

Dashboard

Business Insights

Files Included

How to Use

Author

Markdown Syntax

Heading

Main Heading

Output

Main Heading

Sub Heading

Heading

Smaller Heading

Heading



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Bold Text

****Excel Dashboard****

Output

Excel Dashboard

Italic

Excel

Output

Excel

Bold + Italic

*****Excel*****

Strike Through

~~Old Dashboard~~

Horizontal Line

Creates

Bullet List

- Excel

- SQL

- Power BI

Output

- Excel
- SQL
- Power BI

Numbered List

1. Data Cleaning

2. Dashboard

3. Insights

Check List

- [x] Dashboard Completed

- [x] README Completed

- [] SQL Version Pending



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Output

- [x] Dashboard Completed
- [x] README Completed
- [] SQL Version Pending

Block Quote

> Data drives better decisions.

Output

Data drives better decisions.

Inline Code

``XLOOKUP()``

Output

XLOOKUP()

Code Block

```
```excel
=XLOOKUP()
```
```

Output

=XLOOKUP()

Hyperlinks

Syntax

[Text](URL)

Example

[LinkedIn](https://linkedin.com)

Images

Syntax

![Dashboard](Dashboard.png)

Clickable Image

[![Dashboard](Dashboard.png)](Dashboard.png)

Tables

| Tool | Purpose |
|------|---------|
|------|---------|

| | |
|-------|-------|
| ----- | ----- |
|-------|-------|

| | |
|-------|-----------|
| Excel | Dashboard |
|-------|-----------|

| | |
|-----|-------|
| SQL | Query |
|-----|-------|

| | |
|----------|---------------|
| Power BI | Visualization |
|----------|---------------|

Output



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| Tool | Purpose |
|----------|---------------|
| Excel | Dashboard |
| SQL | Query |
| Power BI | Visualization |

Emoji



Emojis improve readability.

Badges

Example
Excel

SQL

Power BI

Python

(You can later use shields.io badges.)

Repository Sections

A professional Data Analytics README should include:

1. Project Title

Sales Dashboard Analysis

2. About Project

Explain in 4–5 lines.

3. Business Problem

Example

A retail company wants to analyze monthly sales performance.

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4. Objectives

Example

- Find top-selling products
- Analyze regional sales
- Identify low-performing months

5. Dataset

Mention

- Rows
- Columns
- Source
- Time Period

Example

1000 Records

12 Columns

Retail Sales Dataset

6. Tools Used

- Excel
- Power Query
- Pivot Table
- Charts
- Power Pivot

7. Skills Demonstrated

Example

- Data Cleaning
- Pivot Tables
- Dashboard Design
- Lookup Functions
- Conditional Formatting
- Dynamic Arrays

8. Project Workflow

Raw Data



Cleaning



Transformation

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↓
Analysis

↓
Dashboard

↓
Business Insights

9. Dashboard Preview

Insert Screenshot

10. Business Insights

Example

- North Region generated highest revenue.
- Electronics contributed 42% of total sales.
- Revenue increased by 18% in Q4.

11. Files Included

Dashboard.xlsx

Dataset.csv

README.md

Images

Presentation.pdf

12. Learning Outcomes

Example

- Learned Data Cleaning
- Created Dynamic Dashboard
- Used Pivot Tables
- Generated Business Insights

13. Future Improvements

Example

- Add Power BI Dashboard
- SQL Version
- Python Analysis

14. Contact



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GitHub
LinkedIn
Email
Portfolio

README Best Practices

- ✓ Clear project title
- ✓ Business-oriented description
- ✓ High-quality screenshots
- ✓ Professional formatting
- ✓ Proper headings
- ✓ No spelling mistakes
- ✓ Simple language
- ✓ Explain business problem before formulas
- ✓ Mention tools and skills
- ✓ Keep it updated

Common Mistakes

- ✗ No README
- ✗ Only uploading files
- ✗ No screenshots
- ✗ No business explanation
- ✗ Poor formatting
- ✗ Large paragraphs
- ✗ No project objective
- ✗ No contact information

Keyboard Shortcuts (GitHub Markdown Editor)

| Shortcut | Purpose |
|-------------|---------------------------------|
| Ctrl + B | Bold |
| Ctrl + I | Italic |
| Ctrl + K | Insert Link |
| Ctrl + Z | Undo |
| Ctrl + Y | Redo |
| Ctrl + C | Copy |
| Ctrl + V | Paste |
| Ctrl + X | Cut |
| Ctrl + A | Select All |
| Ctrl + S | Save (editor/browser dependent) |
| Tab | Indent list/code |
| Shift + Tab | Outdent |

Example README Structure (Complete)

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Sales Dashboard Analysis

About the Project

This project analyzes retail sales data using Microsoft Excel to identify sales trends, top-performing products, and regional performance.

Business Problem

The company wanted to monitor sales performance and improve decision-making through interactive dashboards.

Dataset

- 1,000 Records
- Retail Sales Dataset
- 12 Columns

Tools Used

- Microsoft Excel
- Pivot Tables
- Pivot Charts
- Power Query
- Conditional Formatting

Skills Demonstrated

- Data Cleaning
- Dynamic Arrays
- Lookup Functions
- Dashboard Design
- Business Analysis

Dashboard Preview

(Insert Dashboard Screenshot)

Key Business Insights

- North region generated the highest revenue.
- Electronics was the top-performing category.
- Q4 showed the highest monthly sales.

Files Included



- Dashboard.xlsx
- Dataset.csv
- README.md
- Dashboard.png

Author

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Data Analytics | Excel | SQL | Power BI | Python

GitHub: <https://github.com/yourusername>

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Interview Question

Q: Why should a Data Analyst maintain a good README?

Answer: A README acts as the documentation of a project. It explains the business problem, dataset, tools, workflow, and insights, making it easy for recruiters and collaborators to understand the project. A professional README increases the credibility of your portfolio and demonstrates both technical and communication skills.



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